

Technical Document #499: Retrieval Augmented Generation - Comprehensive Overview

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Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Hallucination detection identifies when LLMs generate factually incorrect information.
- Semantic search finds documents based on meaning rather than exact keywords.